

ABSTRACT

The insurance industry demands robust, transparent, and efficient systems to manage policies and process claims. This project presents a comprehensive Insurance Management System designed to streamline the entire lifecycle of insurance operations, from user registration and policy acquisition to claim submission and auditing. By digitalizing these core organizational workflows, the system aims to reduce manual overhead, minimize processing delays, and provide a seamless, responsive experience for both clients and internal enterprise staff.

Developed as a modern full-stack web application, the platform integrates a scalable .NET 8 core backend with a dynamic, fast-loading Angular 21 frontend. The system implements a sophisticated role-based access control architecture, accommodating various user personas including Customers, Agents, Claims Officers, and Administrators. A standout technical feature is the custom "least busy" routing algorithm used for the optimal assignment of Agents and Claims Officers to incoming requests, ensuring balanced workloads and rapid resolution times. Furthermore, the application incorporates external cloud integrations, such as Vercel Blob for secure document storage and Quest PDF for the automated, high-fidelity generation of system reports and policy documents.

Ultimately, this system provides an end-to-end operational solution that significantly enhances data transparency and administrative efficiency. Clients can effortlessly navigate available insurance policies, request coverage, and track claim statuses in real time, while officers benefit from centralized management dashboards that facilitate organized, rapid decision-making. By combining a resilient, modern technology stack with intelligent workflow automation, this project successfully delivers a secure and highly scalable platform tailored to meet the rigorous demands of contemporary insurance administration.